Placement in xy-plane

1) Coordinate argument transform

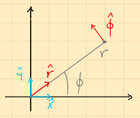
we know $\bar{A}$ is azimuthal $\Rightarrow \bar{A} = A_\phi \cdot \hat{\phi}$,
we want $\bar{A}$ behind the shield $\Rightarrow A^3_\phi \cdot \hat{\phi}$

$$A^3_\phi(r,z) = 2\cdot\mu\cdot a\cdot I\cdot\int_0^\infty d\lambda\left[\frac{c\,\lambda\,z}{z_0^2}\cdot J_1(\lambda a)\cdot J_1(\lambda r)\cdot e(-z_0\cdot z - t(z-z_0))\right]$$

$r \mapsto \sqrt{(x-x_0)^2 + (y-y_0)^2} = \sqrt{x^2+y^2}$

$$A^3_\phi(\sqrt{x^2+y^2},z) = A^3_\phi(x,y,z) = 2\cdot\mu\cdot a\cdot I\cdot\int_0^\infty d\lambda\left[\frac{c\,\lambda\,z}{z_0^2}\cdot J_1(\lambda a)\cdot J_1(\lambda\cdot\sqrt{x^2+y^2})\cdot e(-z_0\cdot z - t(z-z_0))\right]$$

2) Unit vector transform



$$\hat{r} = \cos(\phi)\cdot\hat{x} + \sin(\phi)\cdot\hat{y} = \frac{x}{r}\cdot\hat{x} + \frac{y}{r}\cdot\hat{y}$$

azimuthal $\Rightarrow \hat{r}\perp\hat{\phi}$

$$\hat{\phi} = \cos(\phi+\tfrac{\pi}{2})\hat{x} + \sin(\phi+\tfrac{\pi}{2})\hat{y} = -\sin(\phi)\hat{x} + \cos(\phi)\hat{y} = \frac{-y}{r}\cdot\hat{x} + \frac{x}{r}\cdot\hat{y} = \frac{-y}{\sqrt{x^2+y^2}}\hat{x} + \frac{x}{\sqrt{x^2+y^2}}\cdot\hat{y}$$

We know $\bar{A}^3(x,y,z) = \bar{A}^3_\phi(r,\theta,z)\cdot\hat{\phi}$

$$= \bar{A}^3_\phi(x,y,z)\cdot\hat{\phi}$$

$$= \bar{A}^3_\phi(x,y,z)\cdot\left[\frac{-y}{\sqrt{x^2+y^2}}\cdot\hat{x} + \frac{x}{\sqrt{x^2+y^2}}\hat{y}\right]$$

$$\Phi = \iint_S \bar{B}\,dS = \iint_S \bar{\nabla}\times\bar{A}\,dS = \int_{\partial S}\bar{A}\cdot d\bar{l}$$

$$\bar{l} = x\cdot\hat{x} + y\,\hat{y} = (x_0 + R\cdot\cos\alpha)\hat{x} + (y_0 + R\cdot\sin\alpha)\hat{y}$$

$$dl = R(-\sin\alpha\,\hat{x} + \cos\alpha\,\hat{y})\,d\alpha$$

$$\rightarrow = R\cdot\left(\frac{-y(\alpha)}{r(\alpha)}\cdot\hat{x} + \frac{x(\alpha)}{r(\alpha)}\hat{y}\right)d\alpha$$

$$A^3(x,y,z) = A^3_\phi(r(\alpha),z)\,\hat{\phi} = A^3_\phi\left(\sqrt{x^2(\alpha)+y^2(\alpha)},z\right)\cdot\left(\frac{-y(\alpha)}{r(\alpha)}\cdot\hat{x} + \frac{x(\alpha)}{r(\alpha)}\cdot\hat{y}\right)$$

$$= A^3_\phi(r(\alpha),z)\cdot\frac{R}{r(\alpha)}\cdot\left(y(\alpha)\cdot\sin\alpha + x(\alpha)\cdot\cos\alpha\right)$$

$$= \sim\sim\sim\,\left|(y_0 + R\cdot\sin\alpha)\sin\alpha + (x_0 + R\cos\alpha)\cos\alpha\right|$$

$$= \sim\sim\sim\,\left|y_0\sin\alpha + R\sin^2\alpha + x_0\cos\alpha + R\cos^2\alpha\right|$$

$$= \sim\sim\sim\,\left|(y_0\sin\alpha + x_0\cos\alpha + R\right|$$

$$\Rightarrow \Phi = \int_0^\infty A^3(x,y,z)\,d\alpha = \int_0^\infty A^3_\phi(r(\alpha),z)\cdot R\cdot\frac{(x_0\cos\alpha + y_0\sin\alpha + R)}{\sqrt{(x_0+R\cos\alpha)^2 + (y_0+R\sin\alpha)^2}}\,d\alpha$$

$$= \int_0^\infty\int_0^\infty f(z)\cdot J(r,z)\cdot\frac{R(x_0\cos\alpha + y_0\sin\alpha + R)}{\sqrt{(x_0+R\cos\alpha)^2 + (y_0+R\sin\alpha)^2}}\,d\alpha\,dz$$

RX Loop rotation

1) Expression flux

$$\Phi = \iint_S \bar{B}\cdot d\bar{S} = \iint_S(\bar{\nabla}\times\bar{A})\cdot d\bar{S} = \oint_{\partial S}\bar{A}\cdot d\bar{l}$$) Stokes

2) Parametrization of loop

loop: requirements

* $\bar{u}\perp\bar{v}$, $\|\bar{u}\| = \|\bar{v}\| = 1$
* $\bar{n} = \bar{u}\times\bar{v}$
* ∂S = rx loop = $r(\alpha)$

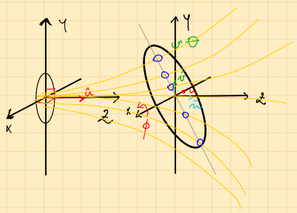
$\rightarrow$ loop: $\bar{r}(\alpha) = c + R\cdot(\cos(\alpha)\cdot\bar{u} + \sin(\alpha)\cdot\bar{v})$

$\downarrow d/d\theta$

$\bar{r}'(\alpha) = R\cdot(-\sin(\alpha)\cdot\bar{u} + \cos(\alpha)\,\bar{v})$

so $\Phi = \int_0^{2\pi} A(\bar{r}(\alpha))\cdot R(-\sin(\alpha)\,\bar{u} + \cos(\alpha)\cdot\bar{v})\,d\alpha$

3) rotation of the loop via rotation about ϕ and γ



$\bar{\bar{R}} = [\bar{u}\ \bar{v}\ \bar{n}]$, bij $\phi = \theta = 0$: $\bar{\bar{R}} = \begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}$

ϕ first then $\theta \Rightarrow \bar{\bar{R}}(\phi,\theta) = \bar{\bar{R}}_y(\theta)\cdot\bar{\bar{R}}_x(\phi)$

$$\bar{\bar{R}}_x(\phi) = \begin{pmatrix}1&0&0\\0&\cos(\phi)&-\sin(\phi)\\0&\sin(\phi)&\cos(\phi)\end{pmatrix}$$

$$\bar{\bar{R}}_y(\theta) = \begin{pmatrix}\cos(\theta)&0&\sin(\theta)\\0&1&0\\-\sin(\theta)&0&\cos(\theta)\end{pmatrix}$$

$$\Rightarrow \bar{\bar{R}} = \begin{pmatrix}\cos\theta & -\sin\theta & 0\\ \cos\phi\sin\theta & \cos\phi\cos\theta & -\sin\phi\\ \sin\phi\sin\theta & \sin\phi\cos\theta & \cos\phi\end{pmatrix}$$

Define the axis and the affine transform / mapping

TX: at origin $\rightarrow O(0,0,0)$ and has axis $\hat{x},\hat{y},\hat{z}$ ← global basis

RX: at arbitrary position $\rightarrow C(x_0,y_0,z_0)$ " " " u,v,n where $n = u\times v$ ← local basis

Define & parametrise the loop

we take RX to be circular, so $p(\alpha) = R\cdot\cos\alpha\cdot\hat{u} + R\cdot\sin\alpha\cdot\hat{v} = \begin{pmatrix}R\cdot\cos\alpha\\R\cdot\sin\alpha\\0\end{pmatrix}$

So we find $p_{global}(\alpha) = C + \bar{\bar{R}}\cdot p(\alpha)$

$$\Rightarrow\begin{pmatrix}x(\alpha)\\y(\alpha)\\z(\alpha)\end{pmatrix} = \begin{pmatrix}x_0\\y_0\\z_0\end{pmatrix} + \begin{pmatrix}u_x & v_x & n_x\\ u_y & v_y & n_y\\ u_z & v_z & u_z\end{pmatrix}\cdot\begin{pmatrix}R\cdot\cos\alpha\\R\cdot\sin\alpha\\0\end{pmatrix}$$

$$d\bar{l}(\alpha) = \frac{d\,\bar{p}_{global}(\alpha)}{d\alpha}\cdot d\alpha = \bar{\bar{R}}\cdot\frac{d\,\bar{p}(\alpha)}{d\alpha}\cdot d\alpha = R\cdot\begin{pmatrix}-u_x\sin\alpha + v_x\cos\alpha\\ -u_y\sin\alpha + v_y\cos\alpha\\ -u_z\sin\alpha + v_z\cos\alpha\end{pmatrix}d\alpha$$

Vectorpotential transform

$$\rho:[0,2\pi]\mapsto R:\alpha\mapsto\sqrt{x^2(\alpha)+y^2(\alpha)}$$

$$A^3 = A^3_\phi(\rho(\alpha),z(\alpha))\cdot\begin{pmatrix}-y(\alpha)/\rho(\alpha)\\ x(\alpha)/\rho(\alpha)\\0\end{pmatrix}$$

Expression flux

$$\Phi = \iint_S\bar{B}\cdot d\bar{S} = \iint_S(\bar{\nabla}\times\bar{A})\cdot d\bar{S} = \oint_{\partial S}\bar{A}\cdot d\bar{l}$$) Stokes

Dot product

$$A^3_\phi\cdot dl = \frac{A^3_\phi}{\rho(\alpha)}\cdot(-y(\alpha)\cdot dl_x + x(\alpha)\cdot dl_y)$$

$$= \frac{A^3_\phi\cdot R}{\rho(\alpha)}\left(-y(\alpha)\cdot(-u_x\sin\alpha + v_x\cos\alpha) + x(\alpha)(-u_y\sin\alpha + v_y\cos\alpha)\right)d\alpha$$

$$= \frac{A^3_\phi\cdot R}{\rho(\alpha)}\cdot\left(-(y_0 + R\cdot u_y\cos\alpha + R\cdot v_y\cdot\sin\alpha)(-u_x\sin\alpha + v_x\cos\alpha) + x_0 + R\cdot u_x\cdot\cos\alpha + R\cdot v_x\cdot\sin\alpha)(-u_y\sin\alpha + v_y\cos\alpha)\right)d\alpha$$

$$= \frac{A^3_\phi\cdot R}{\rho(\alpha)}\cdot\left(y_0(u_x\sin\alpha - v_x\cos\alpha) - x_0(u_y\sin\alpha - v_y\cos\alpha) + R\cdot n_z\right)\cdot d\alpha$$

$$= A^3_\phi\cdot R\cdot\left(y_0(u_x\sin\alpha - v_x\cos\alpha) - x_0(u_y\sin\alpha - v_y\cos\alpha) + R\cdot n_z\right)\Big/\sqrt{(x_0 + R(\cos\alpha\,u_x + \sin\alpha\,v_x))^2 + (y_0 + R(\cos\alpha\,u_y + \sin\alpha\,v_y))^2}\cdot d\alpha$$

* $z(\alpha) = z_0 + R(\cos\alpha\cdot u_z + \sin\alpha\,v_z)$

→ echt de transform zelf